Gawtam Chithra Ramesh

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Education

KTH Royal Institute of Technology — Aug 2024 – Jun 2026
Master of Science in Systems, Controls, and Robotics (GPA 4.62/5.0) — *Stockholm, Sweden*

Indian Institute of Technology Madras — Aug 2019 – Jul 2024
Dual Degree (B.Tech Engineering Design and M.Tech Robotics) (GPA 7.9/10) — *Chennai, India*

- Awarded (among 30 of 800+) the IIT Madras **Young Research Fellowship** based on remarkable research potential.
- Secured **97.64** Percentile in JEE Advanced and **99.10** Percentile in JEE Mains among **1.5M** candidates.
- Secured an All India Rank of **52** out of 50k in Undergraduate Common Entrance Examination for Design.

Skills : Python, C++, C#, MATLAB, ROS2, Unity3D, MuJoCo, PyBullet, PyTorch, OpenCV, YOLO, Docker, Git

Professional Experience

Master's Thesis: Generative Flow Matching for Robot Manipulation — Jan 2026 – Jun 2026
ABB Robotics — *Västerås, Sweden*

- Designed a generative flow-matching planner guided at inference by Signal Temporal Logic robustness gradients.
- Steered the pretrained flow model toward new temporal-logic specifications at inference without retraining.
- Benchmarked on long-horizon reach-avoid navigation against state-of-the-art specification-guided motion planners.

Device Research Intern: 3D Scene Graphs — Jul 2025 – Dec 2025
Ericsson Research — *Lund, Sweden*

- Integrated instance segmentation and tracking into Hydra, a real-time 3D scene-graph construction framework.
- Built a synchronized RGB-D inpainting pipeline that removes privacy-sensitive objects from scene graphs online.
- Benchmarked state-of-the-art inpainting models to balance reconstruction fidelity against real-time latency.

Multi-Object Detection & Tracking Intern — Jun 2021 – Jul 2021
Blurgs Research Labs — *Remote, India*

- Evaluated five state-of-the-art multi-object tracking architectures for an Indian Navy video-surveillance product.
- Trained deep detection-and-tracking models on aerial drone and CCTV footage for real-time multi-target tracking.
- Deployed the top model for online tracking and re-identification of multiple moving targets across video streams.

Research Experience

Taxonomy-Guided VLMs for Deformable Object Manipulation — Jan 2025 – Nov 2025
Prof. Florian T. Pokorny, Robotics, Perception & Learning Lab, KTH — *Stockholm, Sweden*

- Coupled large vision-language models (GPT-4o, Gemini) with a taxonomy to interpret deformable-object scenes.
- Built a prompt-taxonomy framework mapping natural language to robot-parsable structured motion primitives.
- Quantified taxonomy-grounding and natural-language correctness with F1-score and text-similarity metrics.

Autonomous Ultrasound Scan via Structure-from-Motion — Jan 2024 – May 2024
Prof. Nirav Patel, Department of Engineering Design, IIT Madras — *Chennai, India*

- Reconstructed a dense 3D torso point cloud from uncalibrated monocular RGB images via structure-from-motion.
- Densified the reconstruction with multi-view stereo and published the surface as a ROS point-cloud message.
- Segmented the abdominal region and estimated surface normals to plan the probe approach in a ROS/MoveIt pipeline.

Technical Projects

Deep Learning Advanced | DD2610, KTH — Nov 2025 – Jan 2026

- Implemented SimCLR contrastive self-supervised pretraining from scratch with a ResNet18 encoder backbone.
- Reached 86% linear-evaluation accuracy on CIFAR-10 using only 1% of labels with the frozen pretrained encoder.
- Built a Masked Autoencoder ViT from scratch with an asymmetric encoder-decoder and 75% random patch masking.

3D Open-Vocabulary Semantic Mapping | DD2600 Robot Learning, KTH — Aug 2025 – Oct 2025

- Built a CLIP-based semantic voxel-mapping pipeline from RGB-D scans for open-vocabulary object-presence detection.
- Fused open-vocabulary detection with SAM promptable segmentation to ground object instances in a 3D voxel map.
- Trained a ResNet18 presence classifier from scratch, reaching 86.2% validation accuracy and 0.917 ROC-AUC.

Image Analysis & Computer Vision | DD2423, KTH — Oct 2024 – Jan 2025

- Implemented Gaussian and FFT-based image filtering and analyzed denoising behavior across multiple noise models.
- Built a multi-scale differential-geometry edge detector with zero-crossings and a Hough-transform line detector.
- Used SIFT features with RANSAC to estimate homography and the fundamental matrix, then triangulated 3D structure.

Lead Author Publications

- Robust Object-layer Construction of 3DSG Using Instance Segmentation (IMPROVE26)(**Best Paper Award**)
- Synchronized RGB-D Inpainting for Privacy-Aware 3D Scene Graph Construction.(**ICPR26**).(Under Review)
- Scene Understanding in Deformable Object Manipulation via Taxonomy-Guided VLMs.(**IROS25** Workshop)